

AMINO ACID AVAILABILITY AND BRAIN DEVELOPMENT: EFFECTS OF NUTRITIONAL AND METABOLIC INADEQUACIES

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Inadequacies of the brain's amino acid supply are relevant to the processes of protein accretion and transmitter synthesis during brain development. Experimental hyperphenylalaninaemia has demonstrated the consequences of rather severe imbalances of the brain's amino acid supply. An inadequate supply of essential amino acids has been shown to influence a variety of developmental processes, e.g. the rate of cell proliferation, outgrowth of processes, formation of synapses, survival of cells etc. Due to the general interference with normal protein anabolism, the effects are not very specific for the kind of the amino acid imbalance. Different imbalances may have similar consequences. The developmental event that becomes most affected depends primarily on the time of onset and the duration of the imbalance.

The effect of tryptophan on the formation of serotonin demonstrates the consequences of rather mild changes in the supply of individual amino acids during brain development and embryogenesis. Serotonin is an important modulator of normal brain development and acts as a developmental signal in various morphogenetic events. Its production and release by developing serotonergic cells is dependent on their tryptophan supply. Permanent changes of the availability of tryptophan during embryonic life may cause long lasting adaptations of tryptophan and serotonin metabolism.

Consequences of inadequate amino acid availability on protein synthesis and brain development

During brain growth, protein anabolism is greater than protein catabolism. The amino acid requirement is therefore exceptionally high and neuronal protein synthesis is particularly dependent on an adequate supply of at least those amino acids which are essential to brain growth. There is much evidence that short-term as well as moderate variations of the brain's amino acid supply do not affect the process of protein accretion in the developing brain significantly. But what about the very large, and chronic, alterations to the brain's amino acid supply in certain diseases, in inborn errors of metabolism or in protein undernutrition?

An excellent and paradigmatic model to study the influences of such rather severe amino acid imbalances on the developing brain is experimental hyperphenylalaninaemia (hyper-Phe). Suckling rats can be rendered hyperphenylalaninaemic by the daily injection of phenylalanine together with α -methylphenylalanine, an *in vivo* inhibitor of phenylalanine hydroxylase (Greengard, Yoss & Delvalle, 1976). Under these conditions, the blood phenylalanine level is elevated constantly by 20–30 fold, whereas almost all other amino acids are depleted in serum (about 50 per cent of control values) and in the brain (about 70–80 per cent of control values),

(Huether *et al.*, 1984b).

The observation that protein synthesis is disturbed under hyper-Phe conditions has been made by several other groups (Antonias & Coulson, 1975; Geison & Siegel, 1975; Hughes & Johnson, 1976; Hughes & Johnson, 1978a,b). Concomitantly with the decreased incorporation of labelled amino acids, a disaggregation of polyribosomes due to increased RNase activity has been found (Roberts & Morelos, 1976). Supplementation with other large neutral amino acids reversed these effects (Hughes & Johnson, 1978a, 1978b). Clearly, the disturbed protein synthesis is due primarily to the inadequate availability of other amino acids in hyper-Phe.

Two features of this irregular protein synthesis are most remarkable.

(a) The synthesis of the more complex protein species seems to be more affected than that of the simpler, short-chained ones. This was demonstrated elegantly with respect to the various myelin-associated proteins some years ago by Hommes (Hommes, Eller & Taylor, 1982). They found a selective increase in the turnover rate of the smaller myelin proteins in hyper-Phe rats. The rate of synthesis of the larger myelin protein species, however, was decreased, and so was myelin deposition.

(b) In general, protein synthesis of neural cells seems to be most vulnerable to an inadequate supply of amino acids during the phases of most rapid growth of individual brain structures, when net protein anabolism, and therefore the amino acid requirement, is particularly high. In suckling rats the cerebellum is the most rapidly growing brain region. Between day 5 and day 15 post-partum, a massive proliferation of neuroblasts in the external granular layer generates all the granule cells settling down in the internal granular layer. It is the proliferation of these granule cell precursors that is severely affected under hyper-Phe conditions (Huether *et al.*, 1983). Between day 10 and day 20, a huge dendritic tree is grown out by the Purkinje cells, and again,

it is this event of rapid growth which is retarded in hyper-Phe rats (Huether & Neuheff, 1981). The outgrowth of axons is another process requiring massive protein anabolism. The effects of hyper-Phe on this process can easily be monitored in cultured chick spinal ganglia. Rising Phe-concentrations in the medium from 0.1 to 2.0 mM cause a dose-dependent reduction of axonal outgrowth during 24 h. This growth retardation is abolished when other amino acids are supplemented to the medium (unpublished results).

A decreased rate of protein synthesis during these phases of proliferation and outgrowth of processes in the developing hyper-Phe brain is sufficient to affect the tight sequential timing of interdependent neurodevelopmental events, and will therefore result in an increased rate of cell death (Huether, Kaus & Neuheff, 1983). Such deficits in cell number, synaptic contacts and connectivity are irreversible structural changes that limit the further developmental potential of the brain.

Untreated PKU is only one example where the developing brain is confronted with an inadequate availability of amino acids essential to its growth and development. Other inborn errors of amino acid metabolism are characterized by a similar elevation of certain amino acids and a concomitant depletion of other amino acids in the blood and in the brain. Severe protein undernutrition will also restrict the amino acid supply to the developing brain, and in undernourished suckling rats, very similar to hyper-Phe rats, cerebellar development is most affected, and within the cerebellum, similar morphogenetic deficits are found (Huether *et al.*, 1983).

These effects of undernutrition can, of course, be prevented by an increased amino acid supply. The effects of hyper-Phe can also be prevented, when only the depletion of other amino acids in the blood is restored. In dietary-treated PKU-children the depletion of other amino acids disappears when the Phe intake is reduced; it is this effect, rather than the prevention of the accumulation of Phe, that appears to be the reason for the

success of the dietary treatment. In suckling hyper-Phe rats a situation can be created where the depletion of other amino acids in the blood and in the brain is restored, but the Phe-elevation is maintained, simply by the injection of an additional dose of lysine (Huether, Kaus & Neuhoff, 1985). In spite of the persisting elevation of Phe, brain development is then no longer affected. The growth retardation and the morphogenetic deficits are no longer seen, and even the depletion of serotonin and HIAA in the brain is restored (Huether, 1986; Huether *et al.*, 1984a, b).

Consequences of an inadequate amino acid availability on neurotransmitter synthesis and brain development

Even though severe amino acid imbalances can affect protein anabolism and brain development, a possible role of the availability of amino acids as signals involved in the regulation of protein synthesis remains speculative unless future experiments will show that the rate of accretion of certain proteins is selectively affected by an increased or reduced availability of individual amino acids. At present, far more data have been published in support of the idea that the availability of amino acids may be involved in the regulation of developmental events via another mechanism, i.e. by precursor control of the synthesis of important mediators of intercellular communication. Among other 'neurotransmitters', serotonin and catecholamines have been found at early, pre-nervous stages of ontogenetic and phylogenetic development. The interaction of these 'pre-nervous' transmitters with their receptors affects a variety of morphogenetic events during embryogenesis, including the rate of cell division and of protein synthesis. At early phylogenetic stages, e.g. for the free living cells of *Tetrahymena*, the term 'neurotransmitter' does not make much sense in any case. Here, as well as during early ontogenesis, these informational sub-

stances are involved in the control of transcription and translation of genetic information via the activation of second messenger systems and protein phosphorylation, or they modulate other basic cell functions, like contractility, migration and intracellular transport (for references see Huether & Reimer, 1987).

For these reasons, several investigators have proposed that these monoamines may have some pre-transmission (trophic) function prior to their use as mediators of chemical neurotransmission at the synapse (Gustafson & Toneby, 1970; Buznikow & Shmukler, 1980; Lauder, 1985a, b). A summary of such proposed functions of, for example, serotonin is given elsewhere (Huether & Reimer, 1987). Specifically, during neurogenesis and brain development, these monoamines play a part in the regulation of, for example, neural tube closure, neuronal and glial differentiation, cell migration, outgrowth of processes and synaptogenesis (Lauder *et al.*, 1982; Haydon, McCobb & Kater, 1984; Lauder, 1985a, b; Chubakov *et al.*, 1986).

Given this evidence for monoamines as coordinating mediators of intercellular communication during neural ontogeny, the possible control of their formation by the availability of their respective precursor amino acids would be of outstanding importance.

(a) The normally occurring alterations of the availability of tryptophan and tyrosine during embryogenesis and brain development may be part of the control mechanism for appropriate amounts of amines being synthesized and released at appropriate developmental ages by the appropriate cell populations.

(b) Any alteration of the availability of these precursors would interfere with this normal signalling pattern, and may thereby affect normal development.

There is little doubt that the formation of serotonin is affected by the availability of tryptophan, not only in the adult brain but also during early development. In the 2-d old chick embryo, for instance, administration of tryptophan will intensify the serotonin-related fluorescence in the chick

notochord and in the cells of the ventral roof of the neural tube (Wallace, 1982). Similarly, at later developmental stages, an acute injection of tryptophan will cause a prompt elevation not only of brain serotonin, but also of 5-HTP and 5-HIAA (Huether & Reimer, 1987).

These results indicate that the amount of serotonin produced by a certain cell population is dependent on the availability of tryptophan. The availability of tryptophan to the developing embryo is primarily dependent on the amount of tryptophan that can be released from the protein stores of the egg, or in mammals on the maternal tryptophan supply. The availability of tryptophan to individual developing cell populations is dependent on what is left by the others, i.e. on the rate of tryptophan consumption for protein anabolism, of tryptophan metabolism and of tryptophan release by proteolysis in other cells of the developing embryo. An increasing availability of tryptophan to the whole embryo throughout embryogenesis affects this steady state. The injection of additional tryptophan into the unincubated chicken egg does not cause significant changes of concentrations of serotonin and 5-HIAA in the developing brain. However, the ability of the brain of these chick embryos to produce a certain amount of serotonin within a limited time in response to a rise of its tryptophan supply is increased remarkably by this treatment. This increased serotonin-synthesizing capacity is measurable throughout embryogenesis and can still be detected at day 20 after hatching (Huether & Reimer, 1987).

One cannot deny the possibility that shifts in the nutritional supply in a given population will cause shifts in the amount and the proportions of essential substrates (e.g. tryptophan) made available to the descendants by their parents. It is possible that these shifts during early ontogenesis will influence and modify all those processes of metabolic and phenotypic maturation which are dependent on a well defined availability of a particular amino acid.

To test this assumption, rats were bred

for three consecutive generations on a tryptophan-enriched diet (3.8-fold increase in tryptophan content). The concentrations of tryptophan, serotonin and 5-hydroxyindoleacetic acid in the brain of each generation's offspring was measured at days, 5, 10, 15 and 20 post-partum.

In the first generation (offspring of normal rats placed on the tryptophan-enriched diet 2 weeks before mating), the concentration of tryptophan in the brain was significantly reduced until 20-d age (i.e. during the whole suckling period), and so was the brain concentration of serotonin. In the offspring of the F₂ generation a recovery from the depletion of tryptophan and of serotonin in the brain was observed. In the third generation the serotonin concentration in the forebrain was no longer different from controls. However, in this generation, the initial brain weight and the postnatal growth of the body and the brain were significantly decreased. The adult rats of the third generation were significantly smaller than the control rats bred on normal diets, and so was their daily food intake relative to body weight (Thoemke *et al.*, 1983; Huether, 1984).

Conclusions and perspectives

As shown above, an altered amino acid supply can affect normal brain development through its interference with either protein accretion or transmitter formation. The most pressing question is how exactly can we assess, identify and prevent the circumstances under which an altered nutritional supply or a defective peripheral metabolism of amino acid will cause irreversible alterations of normal brain development? There is no simple answer to this question. It took us quite a while to realize that a single nutritional imbalance may have rather different consequences and that its outcome will depend on several other factors.

The first factor is the state of maturity of the developing brain when it is confronted with a particular nutritional imbalance.

There are certain periods during brain development when the brain or, better, the developmental processes that take place in certain areas of the brain, are particularly vulnerable to changes of their nutritional supply. We have learned a lot about the role of the time of onset of nutritional imbalances on the neurologic outcome in children with severe protein malnutrition or in PKU-children whose dietary treatment was started too late.

The second factor interfering with the outcome of nutritional imbalances during brain development is much less understood: it is the interaction of nutritional inadequacies with other extrinsic or intrinsic prerequisites of normal brain development. Some of these extrinsic factors can be other nutrients, which may enhance or suppress the effects of an inadequate supply of certain nutrients on the developing brain, but they can also be other environmental factors, ranging from external temperature and daylight exposure to the intoxication by environmental noxae or the kind and degree of emotional and sensory stimulation or stress. Intrinsic factors that may enhance or suppress the consequences of nutritional imbalances on the developing brain may be individual variations in the efficiency and the state of maturation of the peripheral metabolic pathways and mechanisms involved in the control of the brain's nutrient supply.

And, as if this would make the picture not yet sufficiently complicated, a third factor additionally contributes to the outcome of nutritional imbalances on the developing brain. The role of this important factor is even less understood, because it acts not prior to, not together with, but after the nutritionally-caused defect of the developing brain. This factor is plasticity. Only in rare, exceptional cases we get an impression of the tremendous plasticity exerted by a damaged developing brain. In all other cases we have no idea to what extent plasticity, by means of structural and functional rearrangements, contributes to remodel and to rearrange the brain's connectivity, and to make the

original defect vanish — at least in terms of subsequent brain function. Or, the other way around, without plasticity the effects of nutritional imbalances on the developing human brain may be much more dramatic than they seem to be, from a functional point of view.

The brain's plasticity to compensate for certain developmental defects creates an additional problem which is of equal theoretical and practical importance, and which is related to the reversibility and irreversibility of developmental processes in general, and of developmental failures in particular. If the proper formation of a certain structure or a specific neuronal pathway is disturbed, we end up with structural aberrations, such as deficits in cell number and connectivity. All structural deficits of the developing brain are irreversible in later life. If, however, other neurons, by increasing their dendritic arborization or by additional axonal sprouting and synapse formation will compensate for the original defects, we may not see any functional deficit in later life. The conclusion that the consequences of early nutritional deprivation, for example, can be reversed in later life is wrong. There is no way to reverse any process during brain development, once it has become manifest in terms of structural changes. It is the change of structure that, on the one hand, represents an irreversible final stage of a developmental process and that, on the other hand, becomes the starting point, the basis of all future developmental events.

In order to differentiate between primary causes and secondary consequences of nutritionally induced developmental failures, one has to extrapolate from findings that have been made in animal models designed to investigate the effects of well defined, for example amino acid, imbalances on the developing brain. Nevertheless, one has to be aware that these extrapolations may easily overemphasize the actual contribution of an individual factor, particularly if the original findings were obtained under well described ex-

perimental conditions which exaggerate the role of specific factors, and which do not apply to other, more complex, situations. These are some of the most pressing

problems we are confronted with in our search for consequences of an altered amino acid availability during brain development.

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